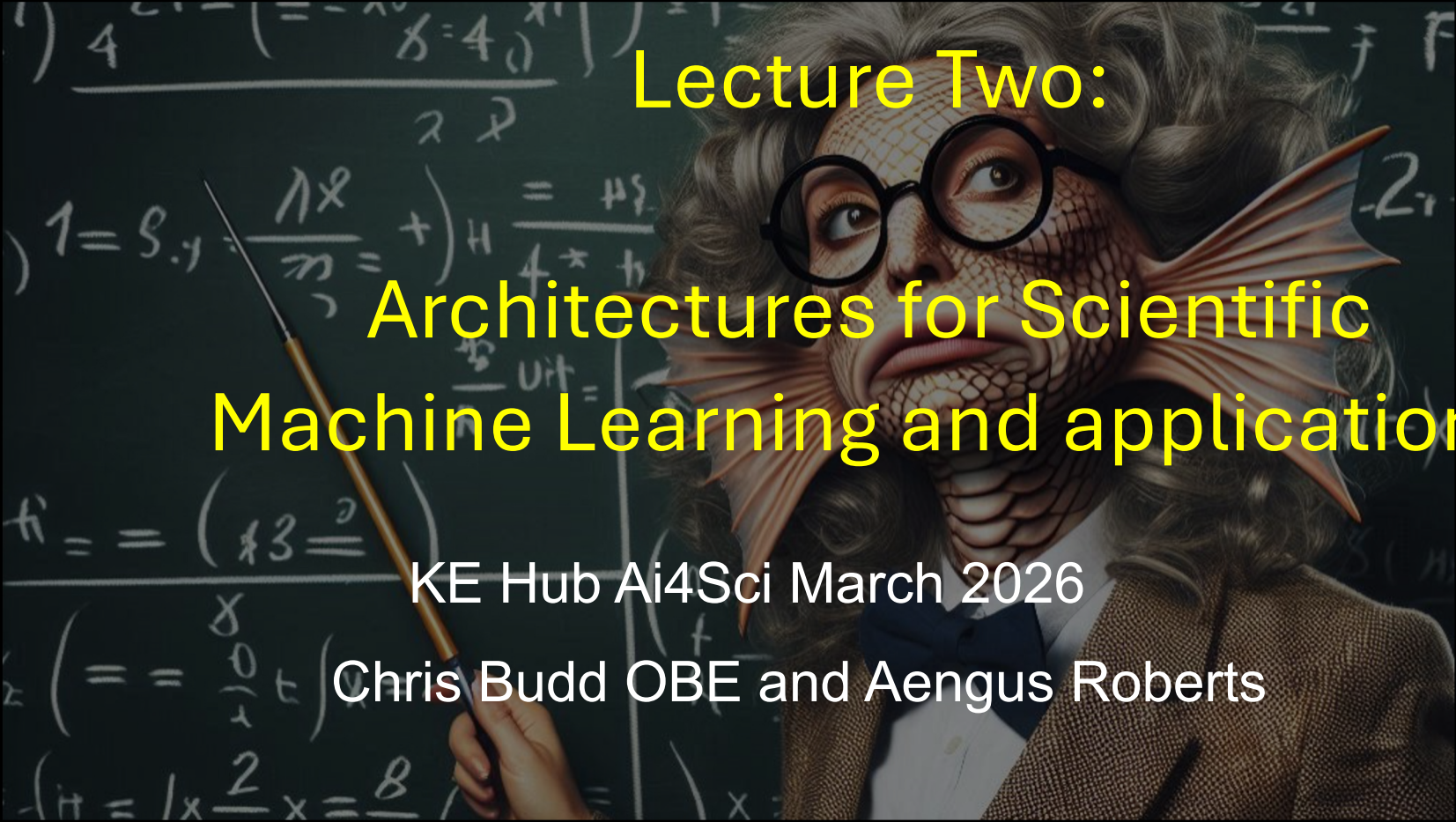




Lecture Two: Architectures for Scientific Machine Learning and applications

KE Hub Ai4Sci March 2026

Chris Budd OBE and Aengus Roberts



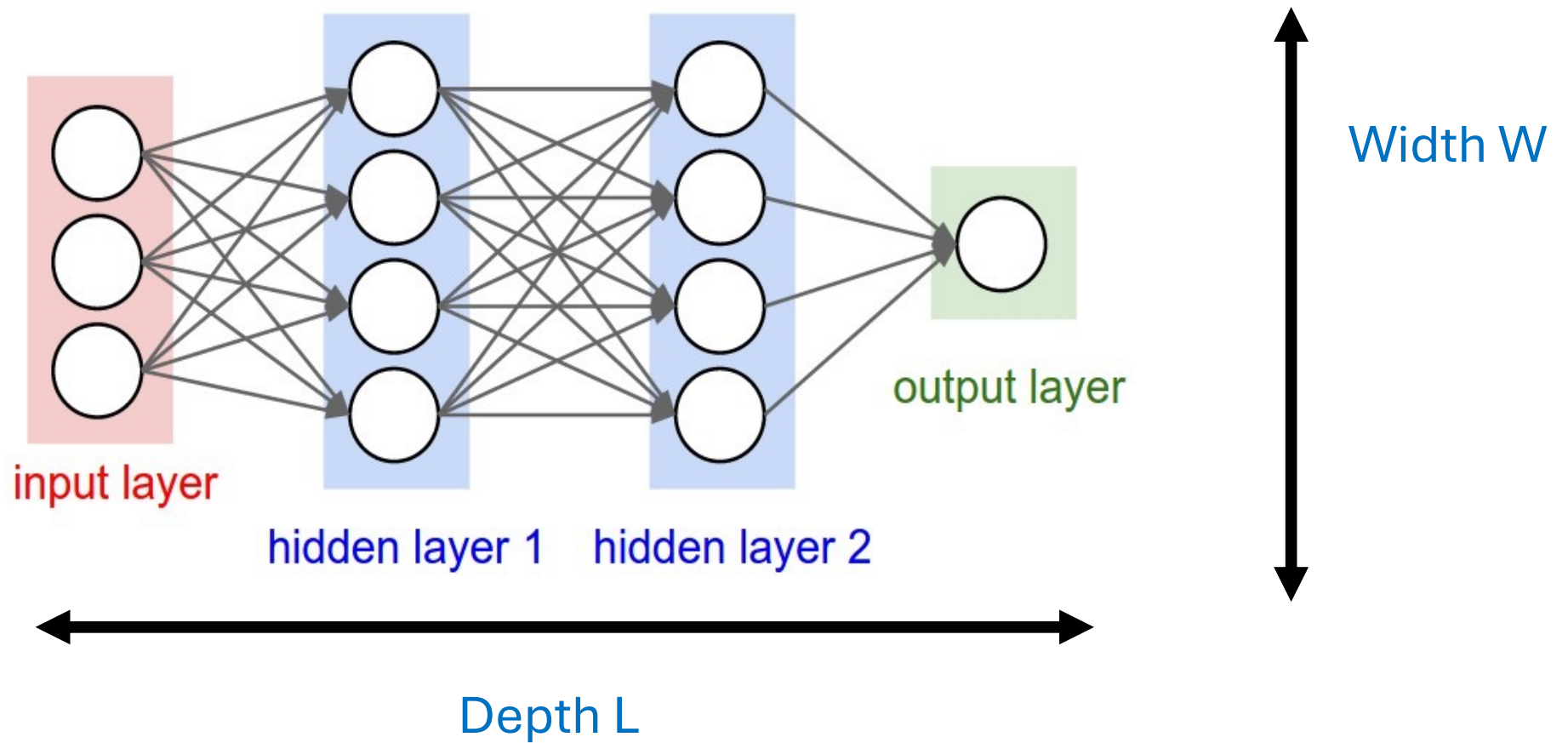
**Knowledge
Exchange Hub**
for Mathematical
Sciences



UNIVERSITY OF
BATH

Multi-Layer Perceptron

Good for classification problems and general use



Applications:

- Classification of data without any special structure
- Approximation of functions (especially in higher dimensions)
- Solution of differential equations using PINNS
- Variational methods

Great ‘expressivity’

‘Optimal error’ scales as: W^{-n} and $e^{-\alpha L}$

Special architectures:

Convolutional Neural Network (CNN)

Uses structured sparse linear operators designed to take a convolution of an image.

Excellent for feature extraction, **image processing**,

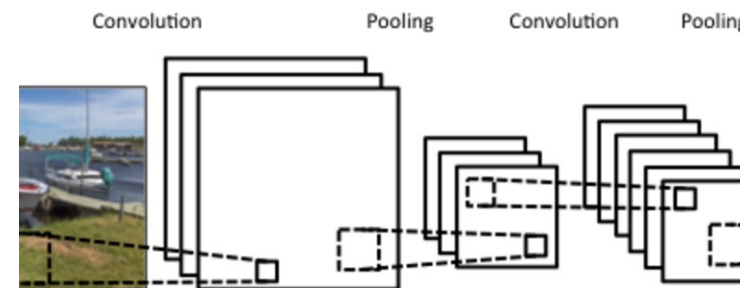
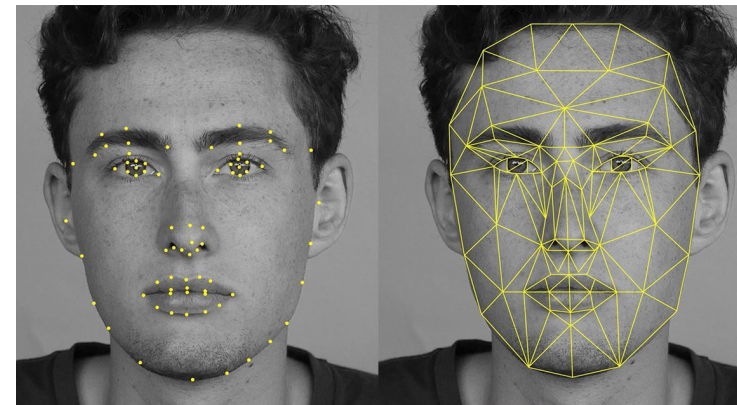
35	40	41	45	50
40	40	42	46	52
42	46	50	55	55
48	52	56	58	60
56	60	65	70	75

 $\times$

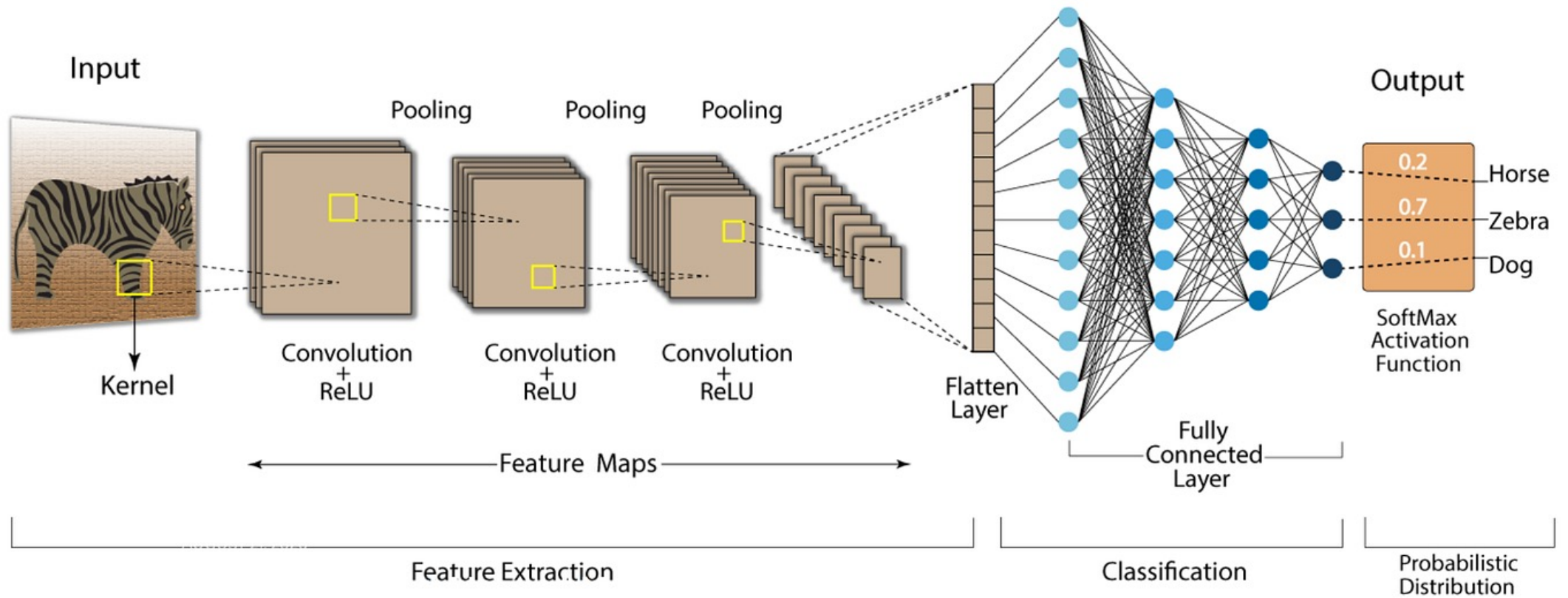
	0	1	0	
	0	0	0	
	0	0	0	

 $=$

		42		



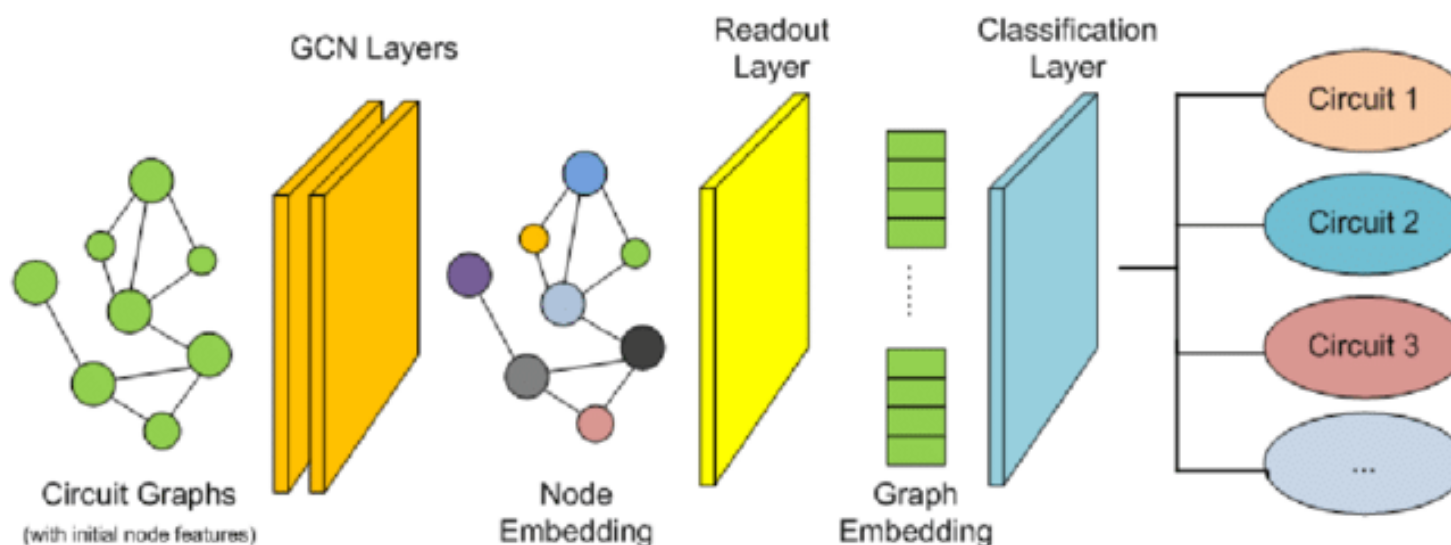
Convolution Neural Network (CNN)



Graph neural networks (GNNs)

Specialized NNs that are designed for tasks whose inputs are [graphs](#)

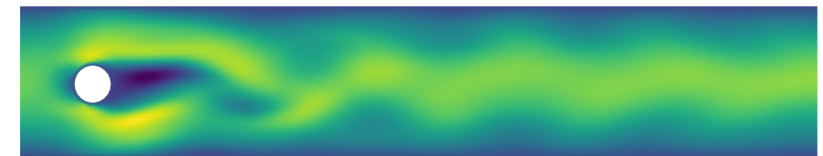
Similar to CNNs. Can be used for structured images, or PDES



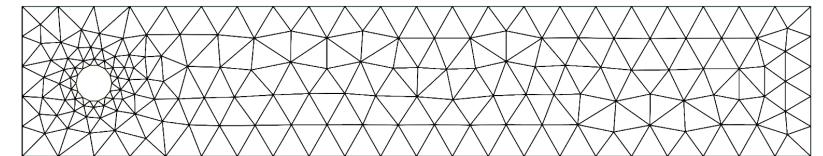
Applications:

They are used across various fields, including:

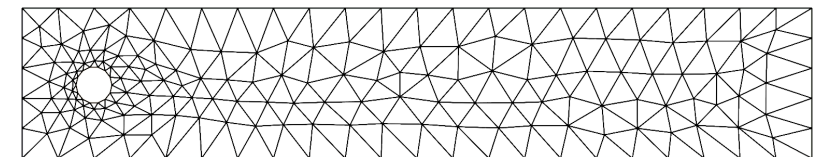
- **Molecular drug design:** Predicting the efficacy of molecules based on their atomic structure.
- **Social network analysis:** Identifying communities and influential users.
- **Physics and engineering:** Modeling unstructured grid data and optimizing mesh relocation in finite element methods (FEMs).
- **Healthcare:** Identifying progression pathways for diseases like Alzheimer's from patient record



Solution velocity field (x-component)



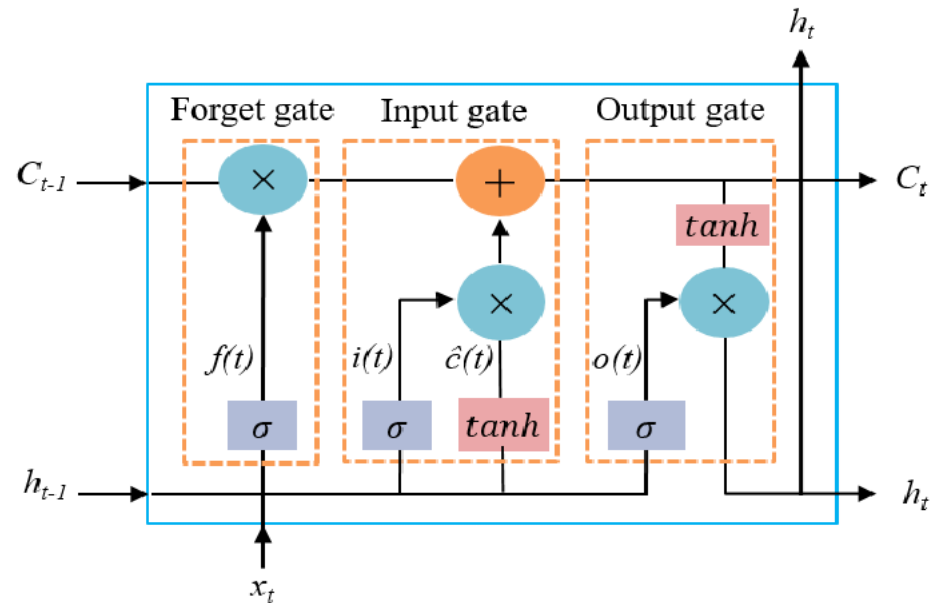
Undeformed mesh



G-Adaptivity deformed mesh (23.57% ER in 32ms)

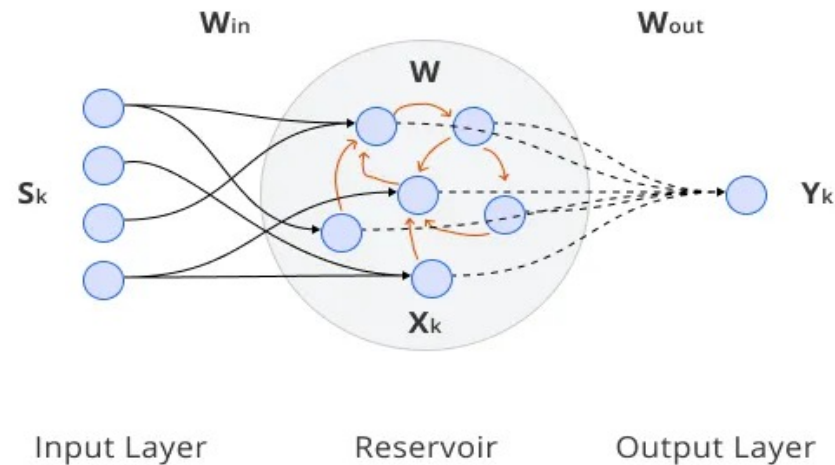
LSTM (Long, Short, Term Memory) Networks

The Long Short-Term Memory (LSTM) architecture is a type of recurrent neural network (RNN) designed to overcome the vanishing gradient problem and effectively **model long-term dependencies in sequential data**. Eg. **short term prediction**





Echo State Networks



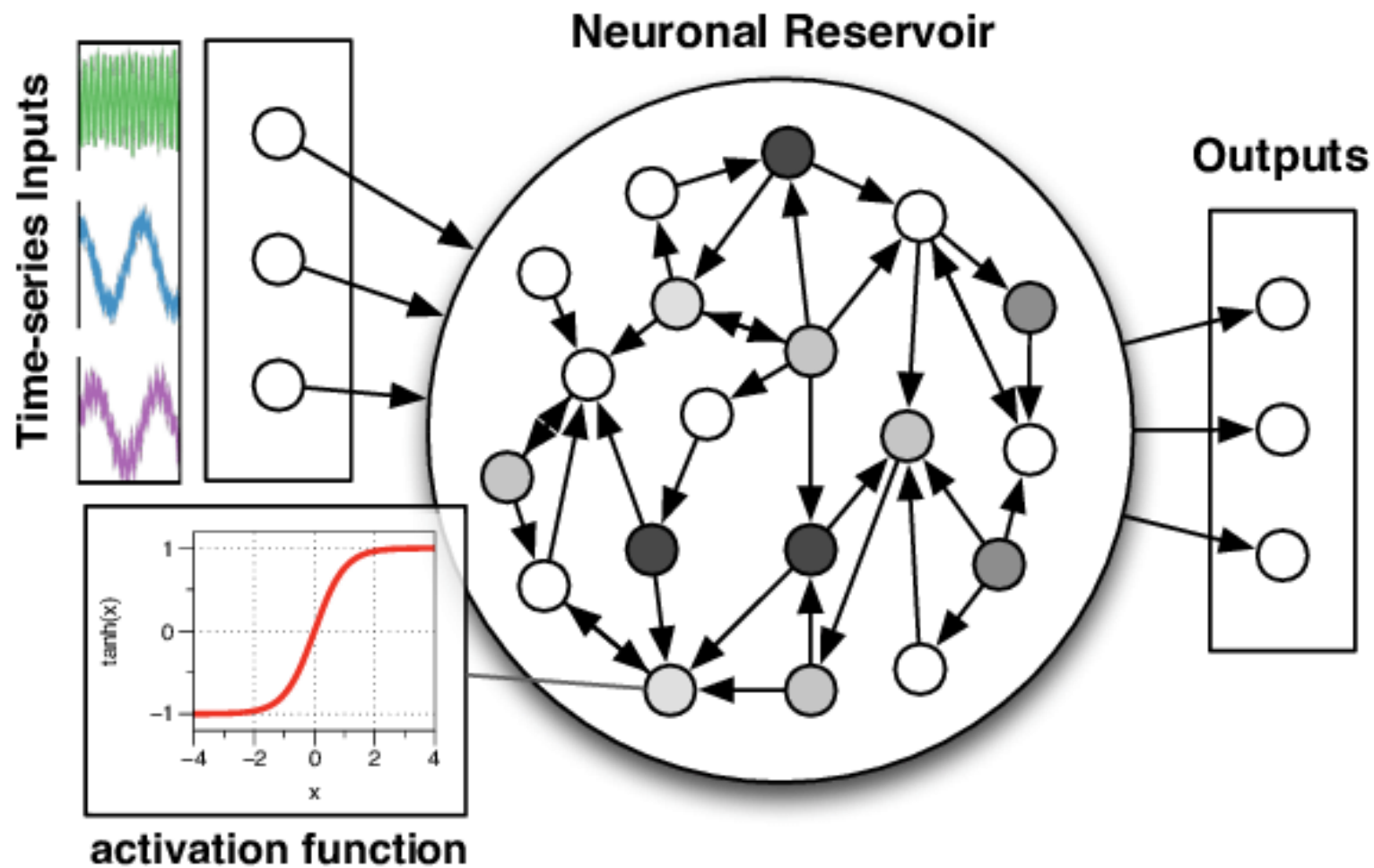
Random (fixed) interior layers

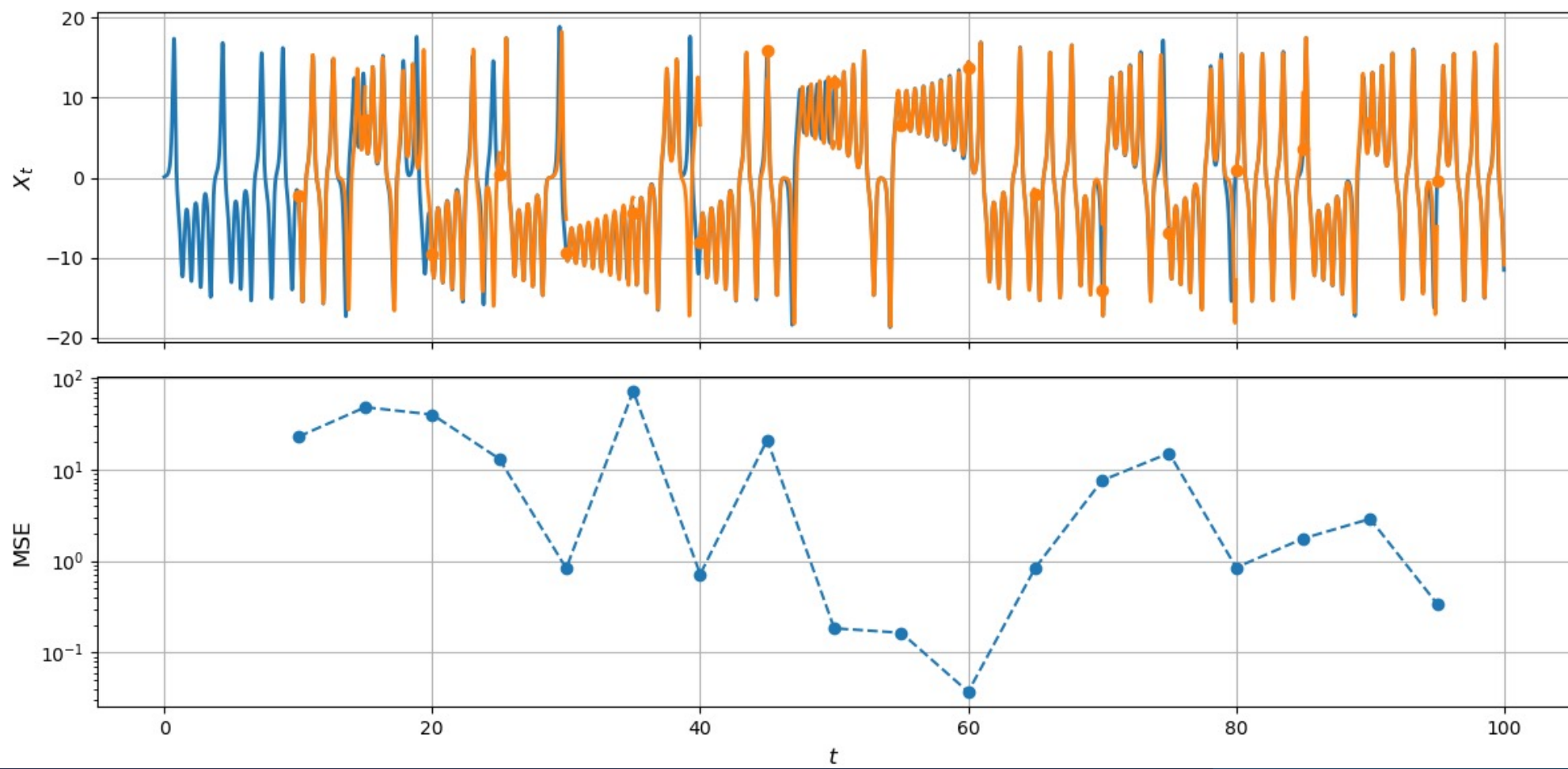
Trainable (linear) OUTPUT layer

Used to make predictions on chaotic systems

Large Language Models/Transformers

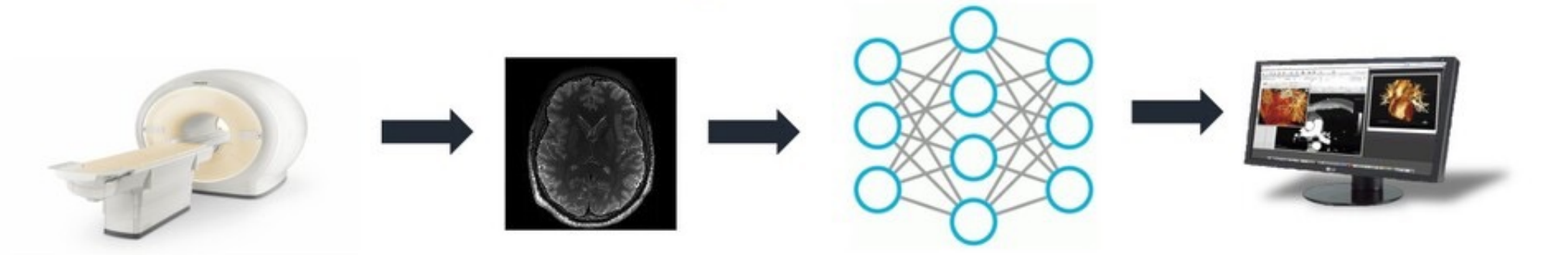
Very powerful .. But not on the course!



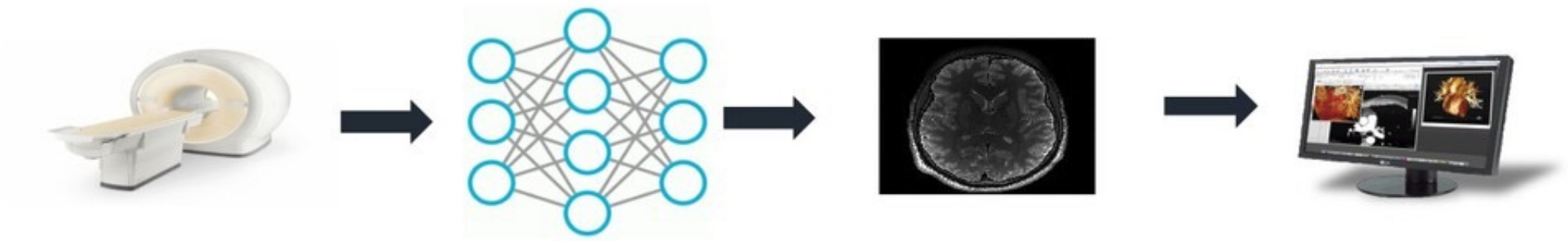


Deep Learning for Inverse Problems

Diagnosis & analysis

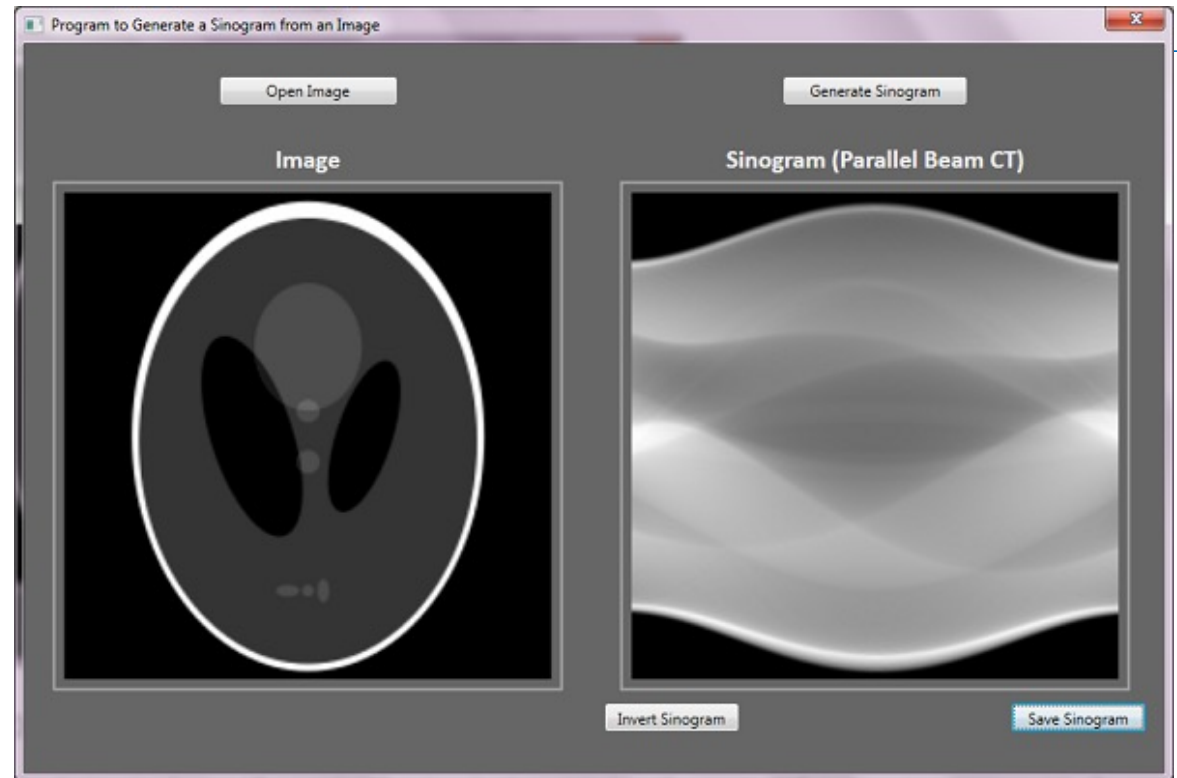


New trend of deep learning: **inverse problems**



$$Ax = y + \text{noise}$$

A: ill-conditioned



Use

$$x = \operatorname{argmin}_x |A x - y|^2 + R(x) : \quad \text{Data error} + \text{Regulariser}$$

$R(x)$

Q: Best regulariser

Tychonov: $R(x) = |B(x)|^2$. TV: $R(x) = |\nabla x|$

Learned: Train the regulariser on a set of problems



Low-dose CT with FBP



Standard Radiation Dose CT



Low Dose CT with DLR

AI and Cultural Heritage: Investigating Artworks and Manuscripts



The Adoration of the Kings (NG592), tempera on wood, about 1472, at the National Gallery, London. Attributed to Sandro Botticelli and Filippino Lippi. Image © The National Gallery, London.

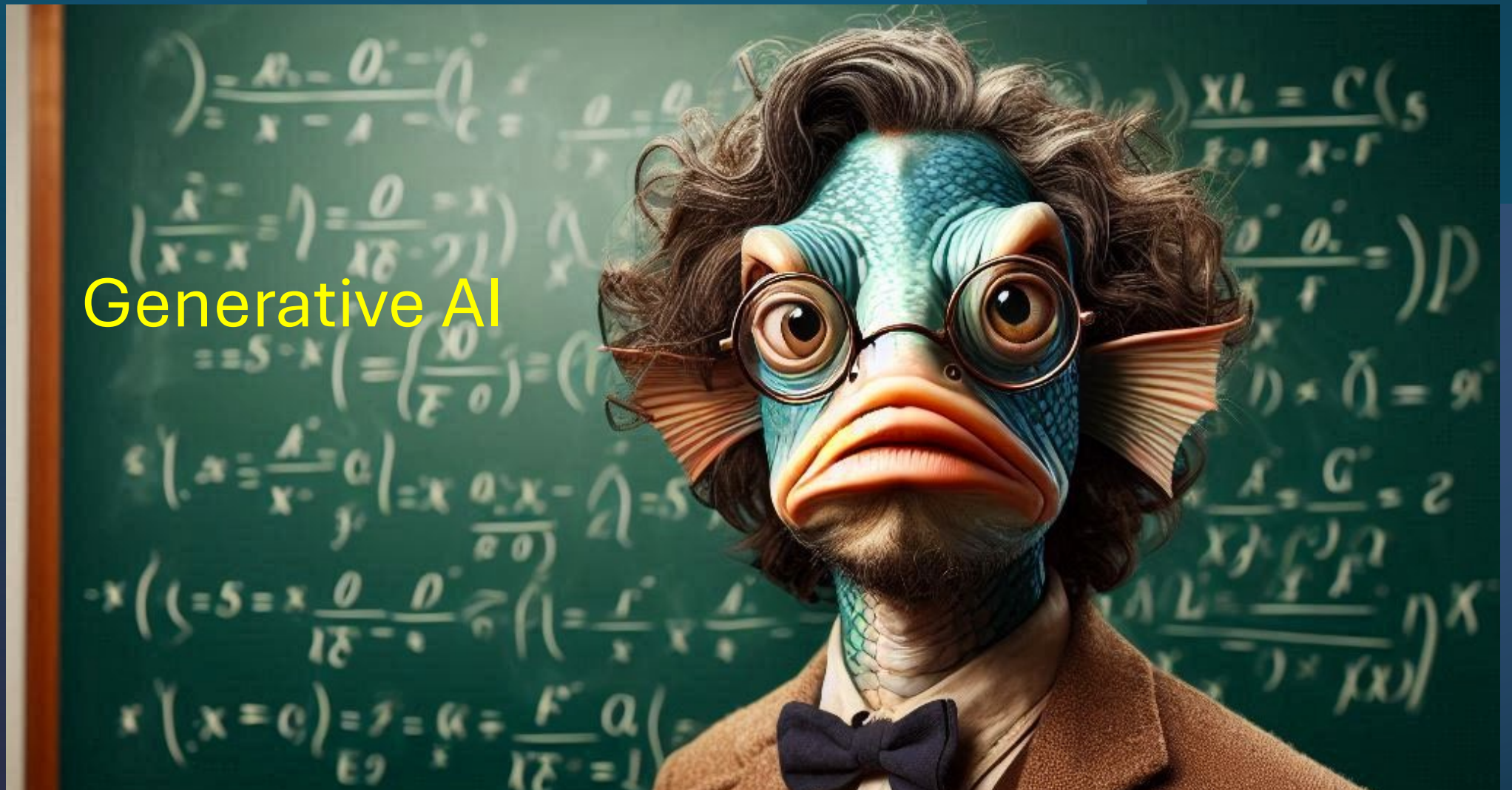
[1] W. Peaslee, L. Wrapson, C.-B. Schönlieb, "Domain generalization and punch mark classification," *Journal of Cultural Heritage*, vol. 72, 2025

[2] M. Zulich, et al., "An Artificial Intelligence System for Automatic Recognition of Punches in Fourteenth-Century Panel Painting," *IEEE Access*, vol. 11, 2023.



Left: Punch marks (from [2]). Right: *St Augustine* (accession number 552) attributed to Simone Martini at The Fitzwilliam Museum, Cambridge; Image © The Fitzwilliam Museum, University of Cambridge.

Generative AI



Generative AI: Image creation

Generate samples that are similar to training examples

Probability distribution of images and classifiers

Take samples from this

Image label/Partial Image

Eg. Fishy mathematician

Trained NN

Generate new samples conditional on the text prompt



Problems with averaging if not done carefully



https://en.wikipedia.org/wiki/Stable_Diffusion

Applications

Inpainting

Image reconstruction

Down scaling

Fake image creation (random)

Fake image creation (labeled by text prompt)

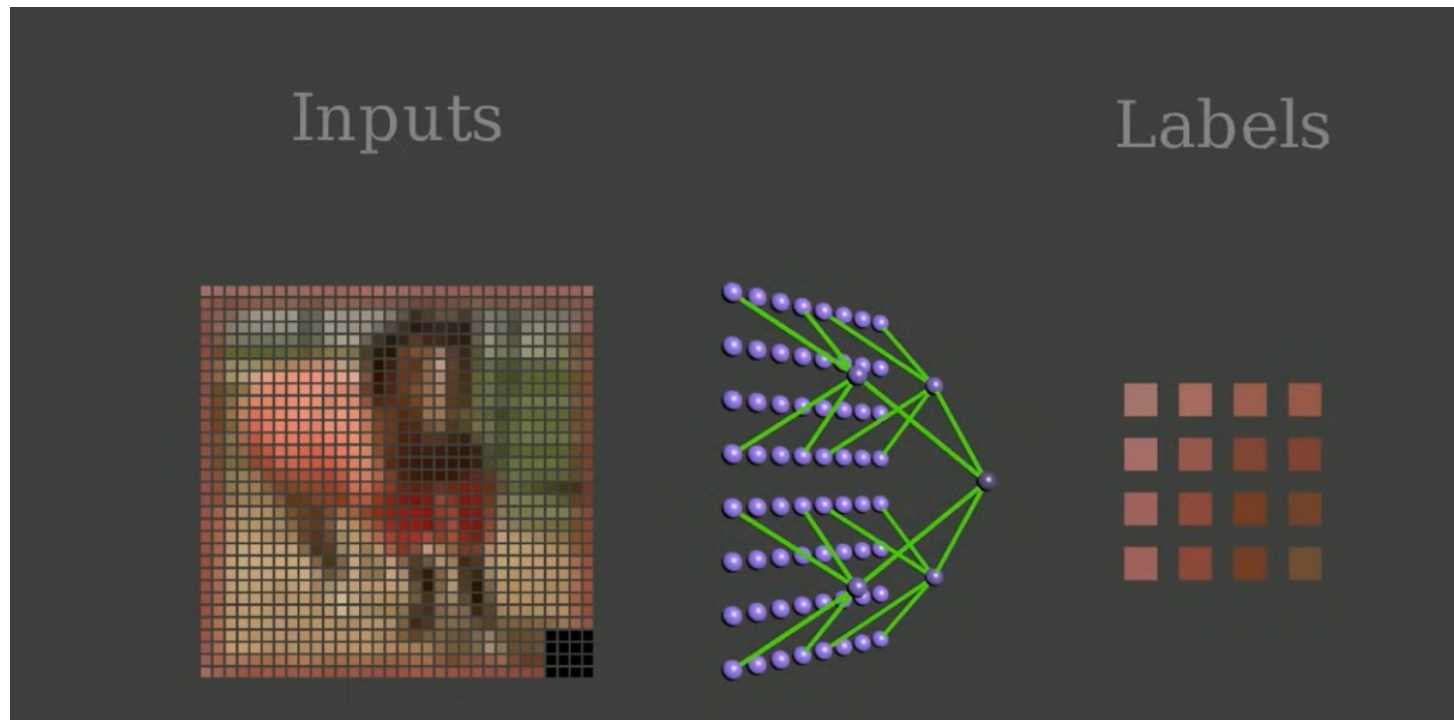
Protein synthesis

Climate prediction



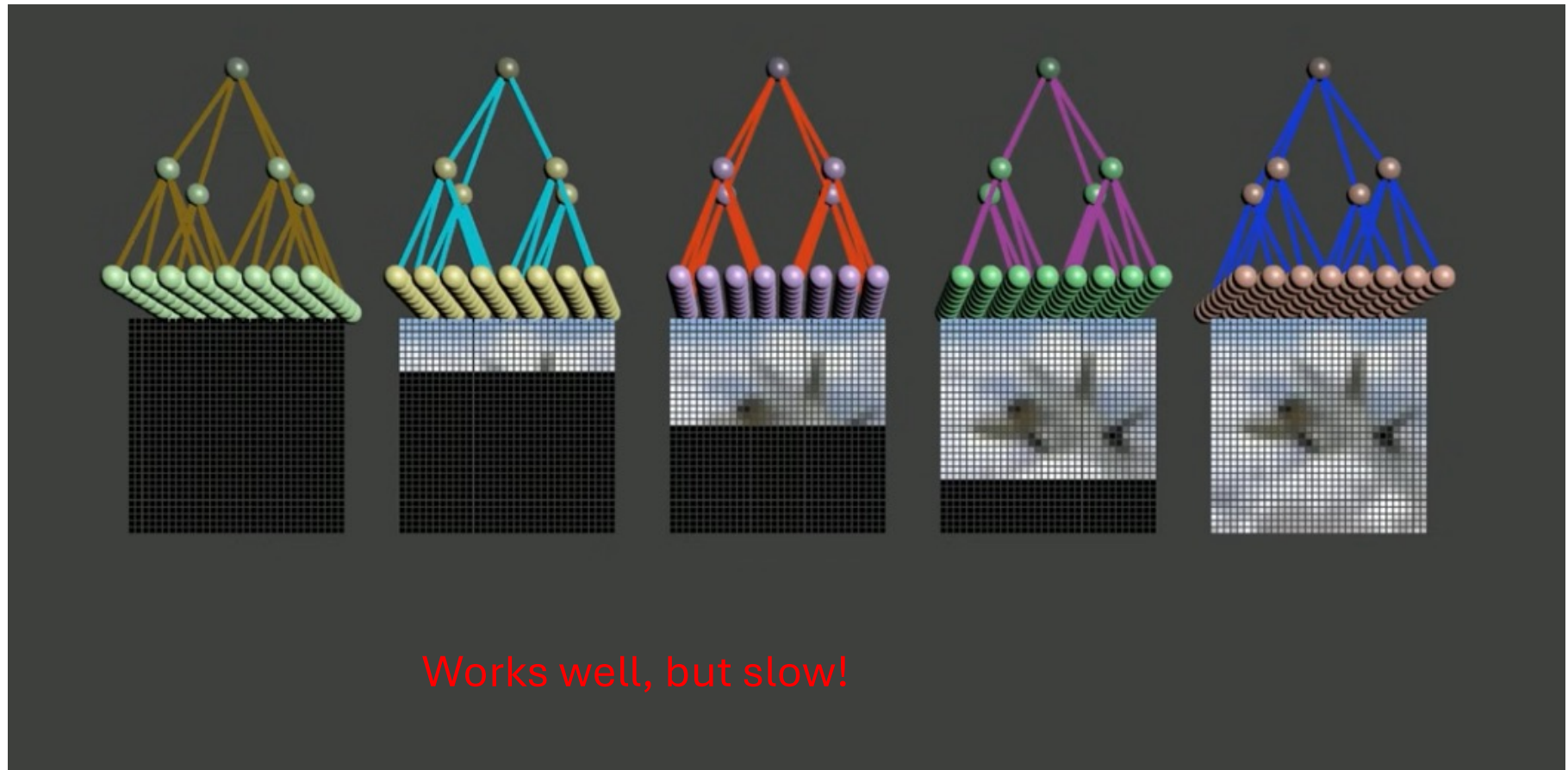
Generative AI Methods I : Auto Regressive Methods [1927!]

(used for example in Chat GPT)

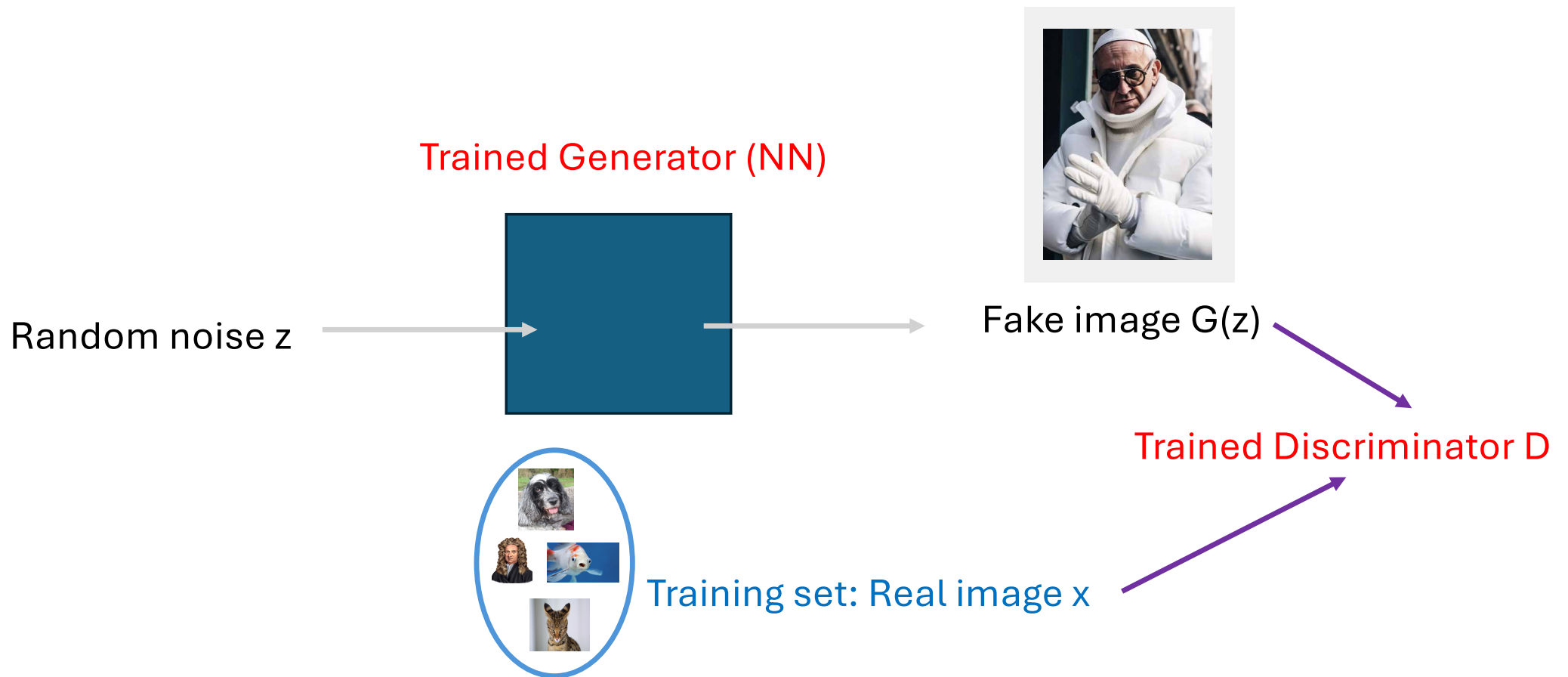


- Given an image with some **Pixels missing**,
- **Train a NN to fill in the missing Pixels** (as a probability distribution)

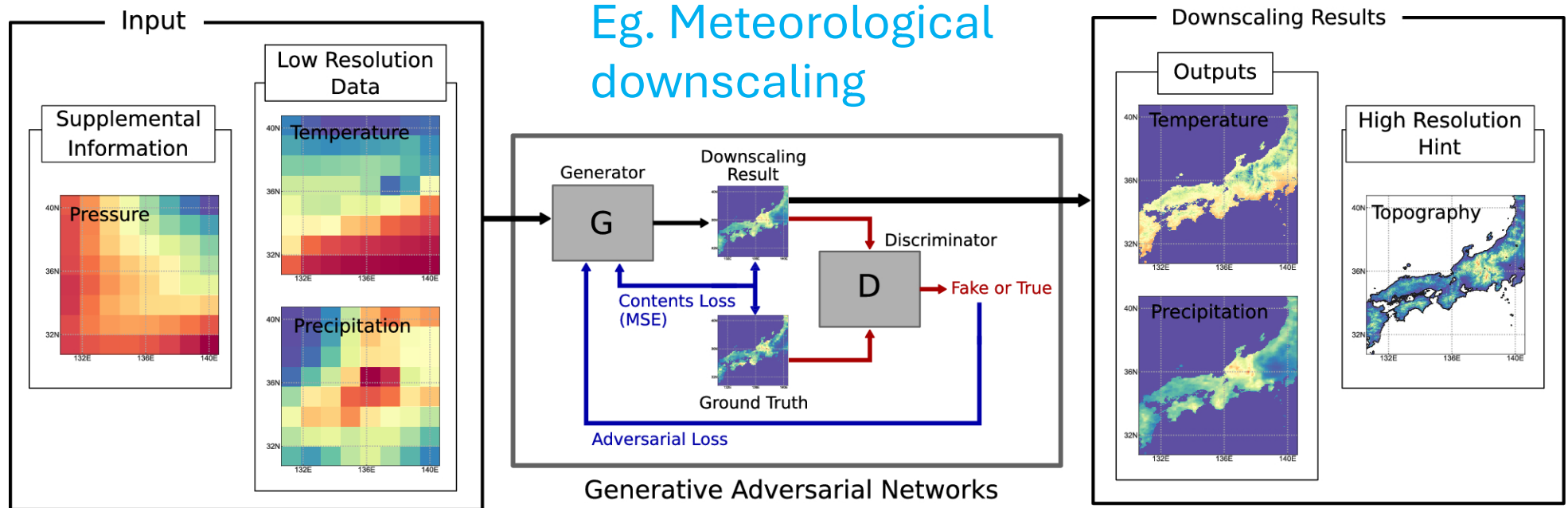
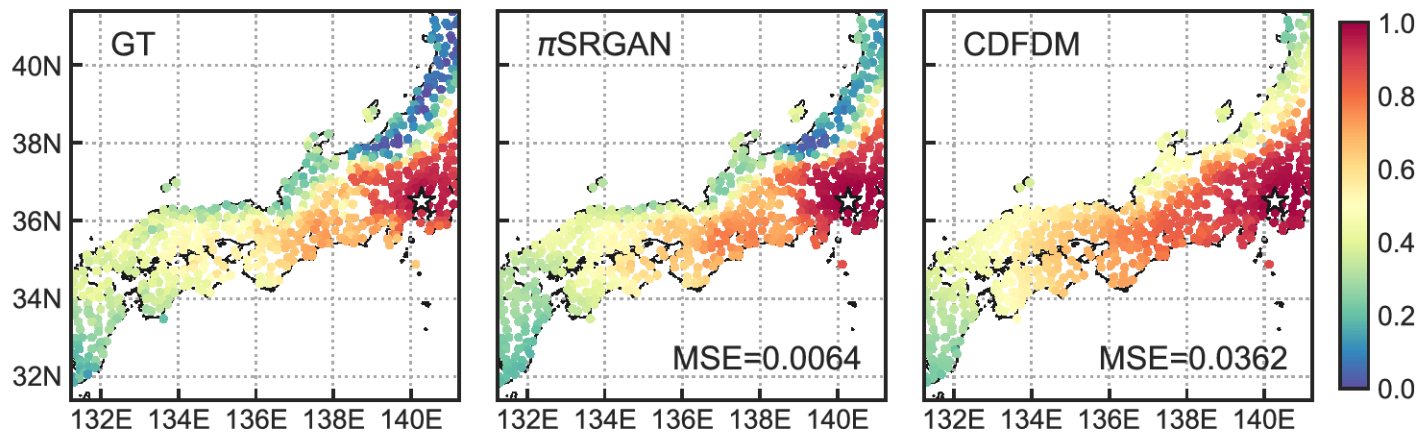
- Do this repeatedly to build up a whole image from an initially blank screen



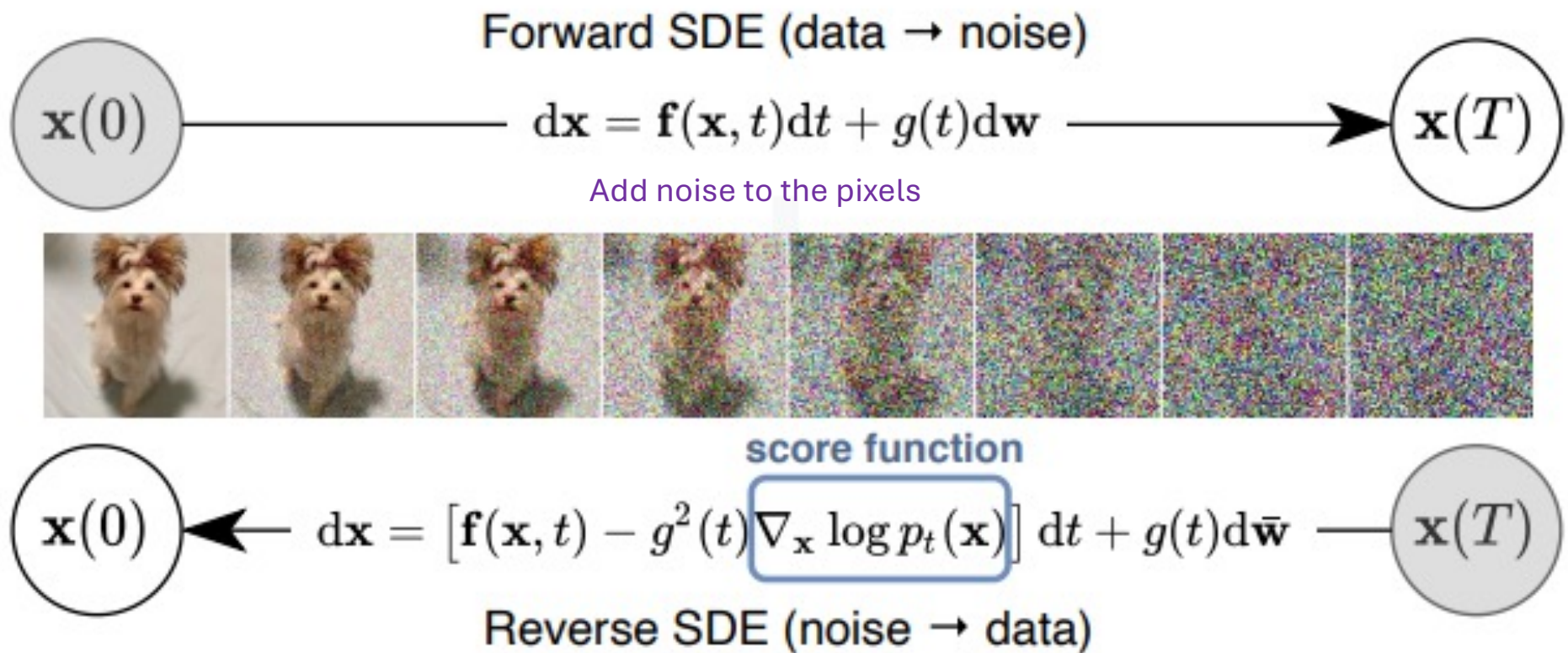
Generative AI Methods II: GANs



Optimise G and D via: $\min_G \max_D E_x(\log(D)) + E_z(\log(1 - D(G(z))))$

A**B**

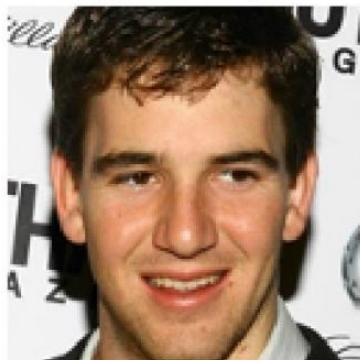
Generative AI Methods III: Diffusion methods (SBD)



MUCH FASTER!!

Eg. Stable diffusion to generate image from a label

Original image x



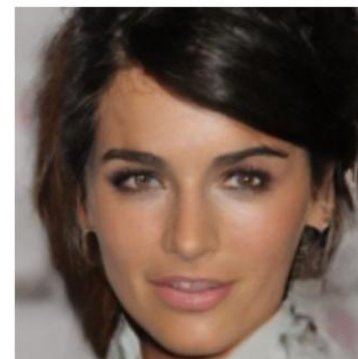
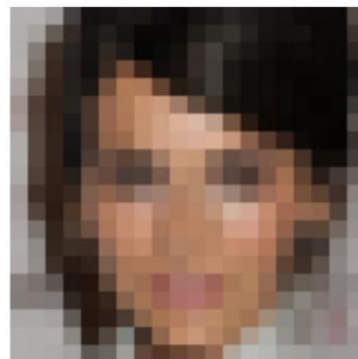
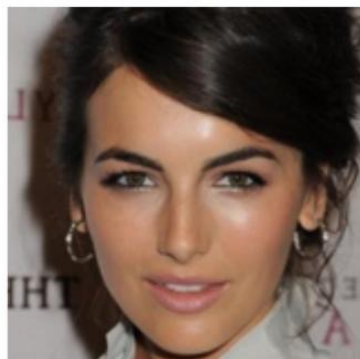
Observation y



Sample from $p_{\theta}(x|y)$



[Schoenlieb et al]



Can produce photo-realistic images .. How can we tell whether they are faked?



Using Maths to Detect **Inpainted** AI Images

Original Image



Fast
Fourier
Transform
Algorithm

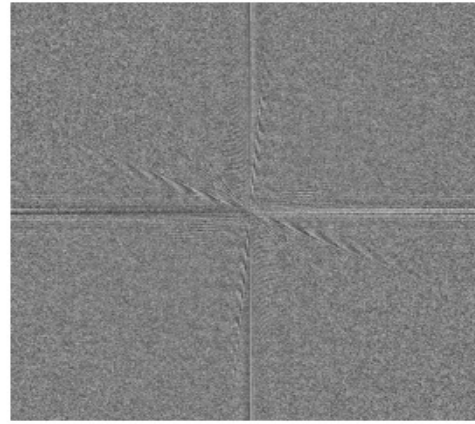


Identifies
changes
in
frequency
at each
image
pixel

Inpainted Image

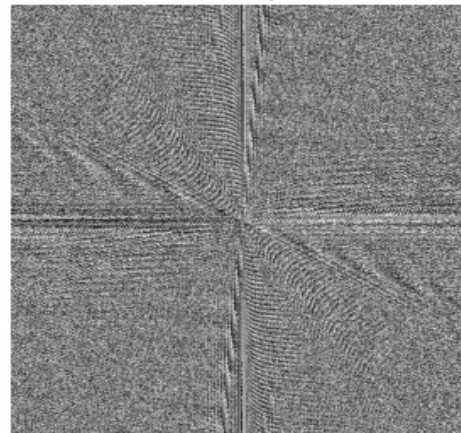


FFT Phase Spectrum



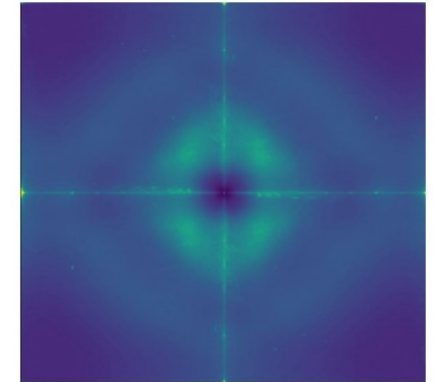
Smooth
textures; no
in-painting

FFT Phase Spectrum

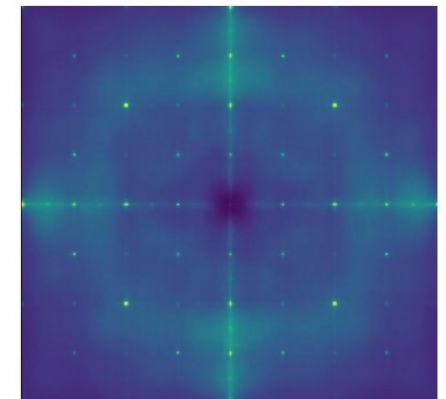


Rough
textures
due to
mask
applied;
indicate
in-painting

**Artefacts to
identify an AI-
generated image**



Real Image Fingerprint



SD Image Fingerprint